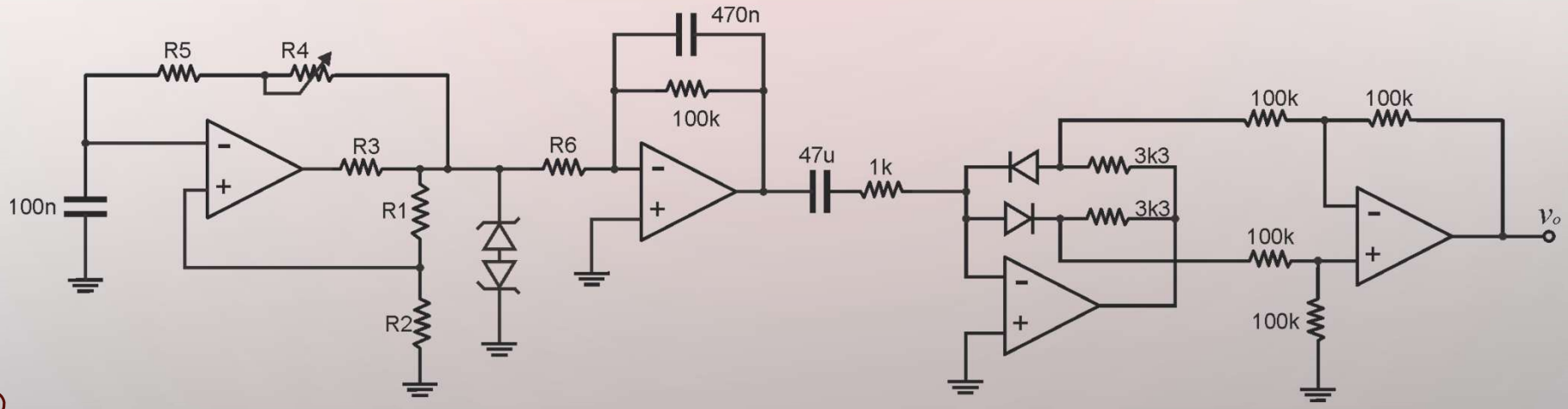
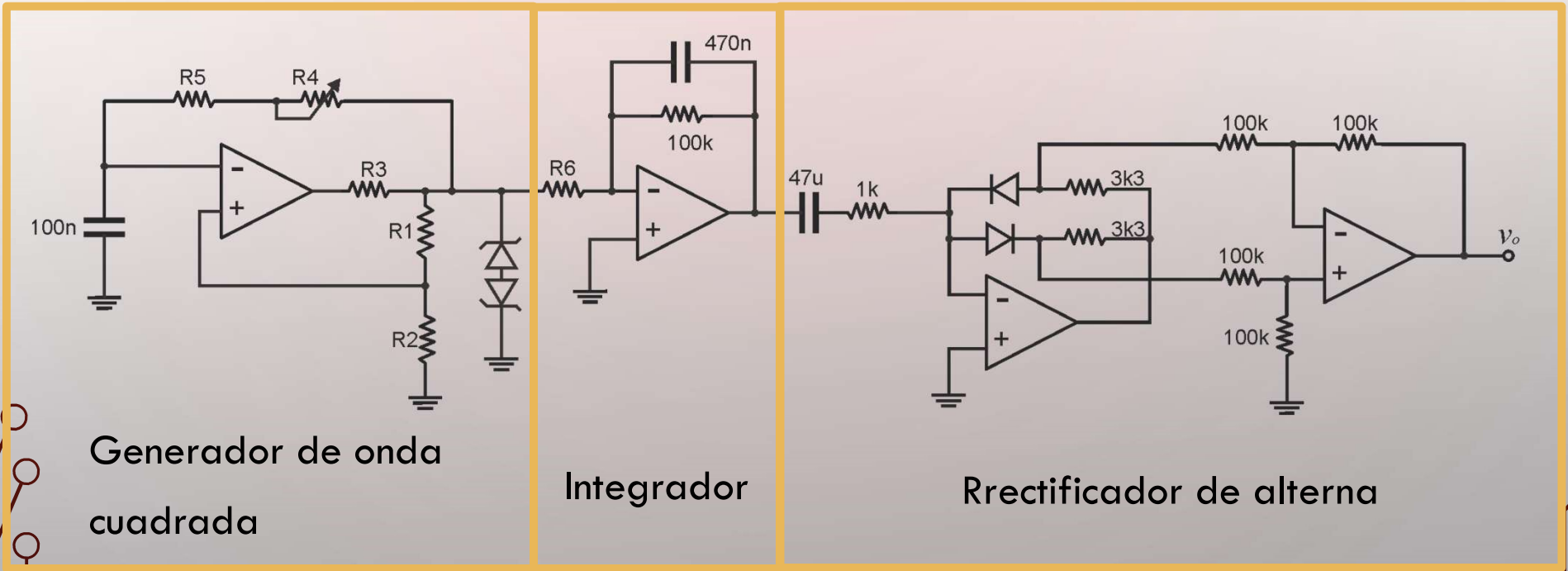
LABORATORIO N°3

AMPLIFICADORES OPERACIONALES

GENERADOR DE ONDA



GENERADOR DE ONDA

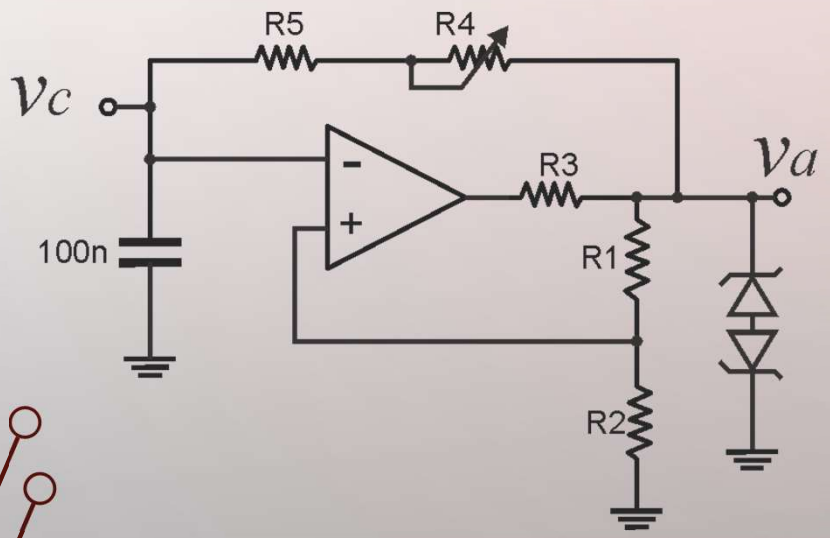


Generador de onda cuadrada

Integrador

Rrectificador de alterna

GENERADOR DE ONDA CUADRADA

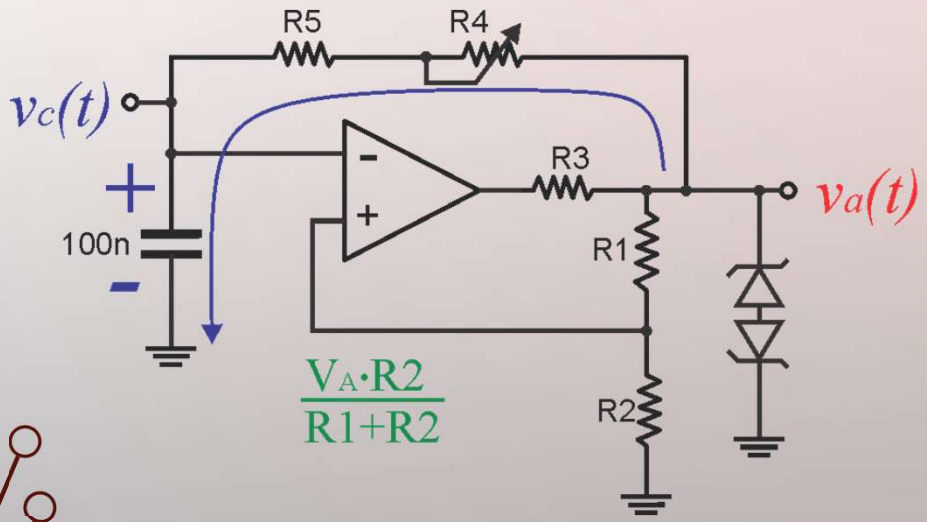


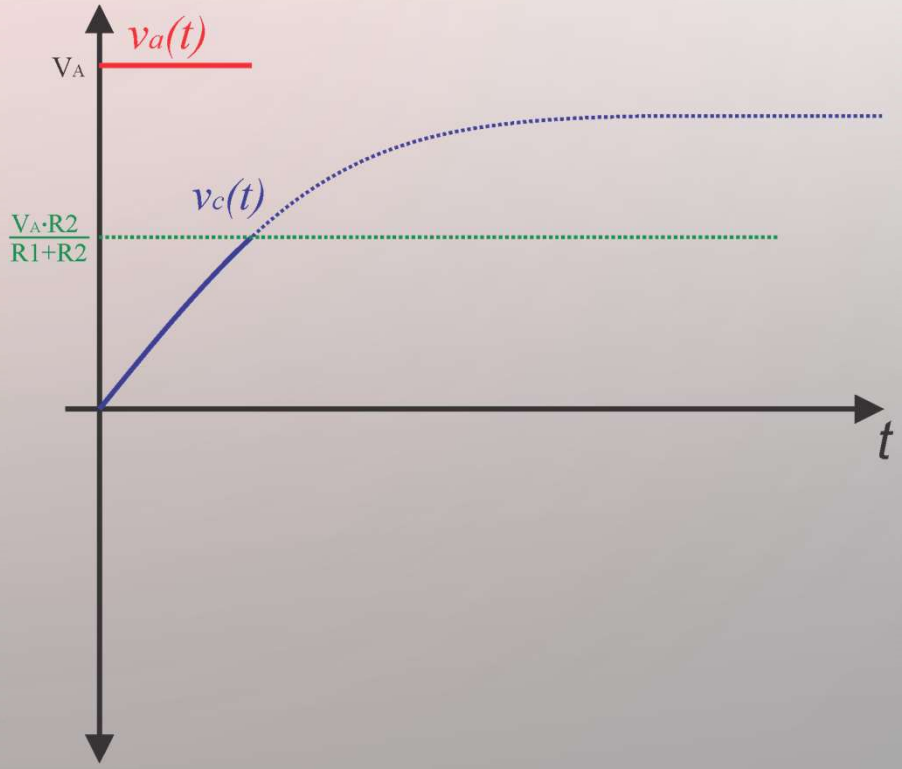
$$\widehat{v_a} = \frac{V_Z + V_\gamma}{2} = 7{,}4V \quad \Rightarrow \quad V_Z = 6{,}8V$$

Adoptamos: $R_1 = R_2 = 10\text{k}\Omega \quad \Rightarrow \quad \widehat{v_c} = \frac{V_Z + V_\gamma}{2} = 3{,}7V$

$$\frac{Vcc - \widehat{v_a}}{\mathrm{R}_3} = \mathrm{I}_\mathrm{o} \quad \Rightarrow \quad \mathrm{R}_3 = 1\text{k}\Omega$$

GENERADOR DE ONDA CUADRADA





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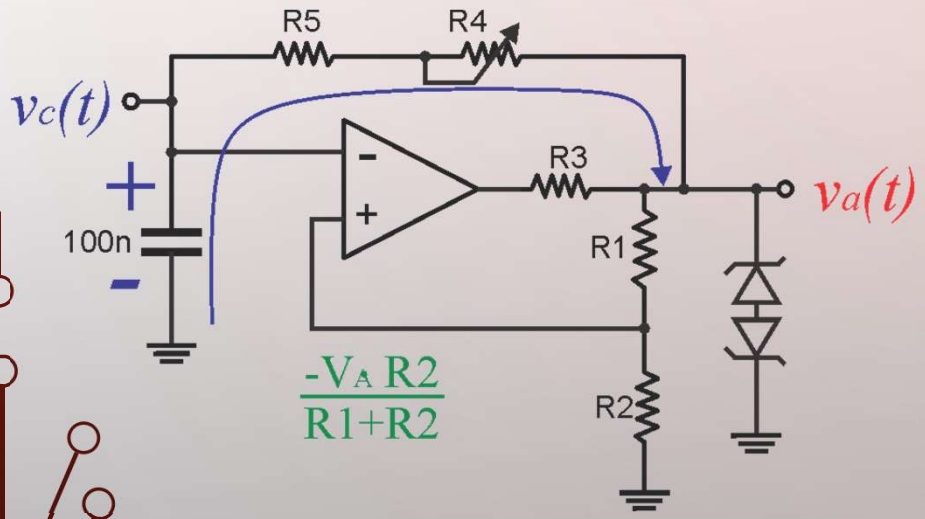
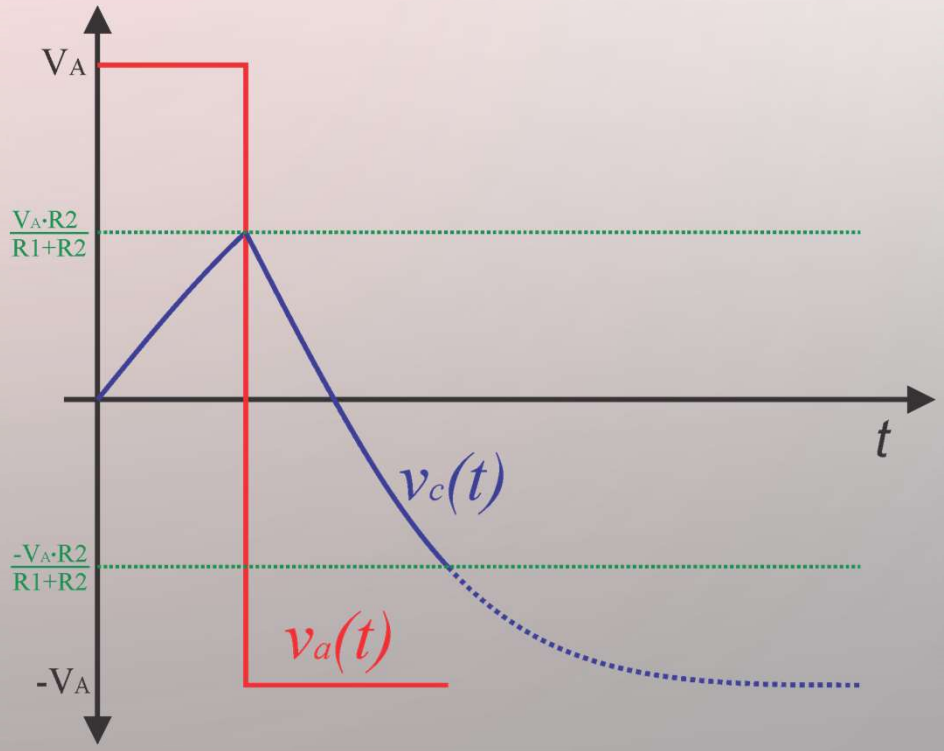
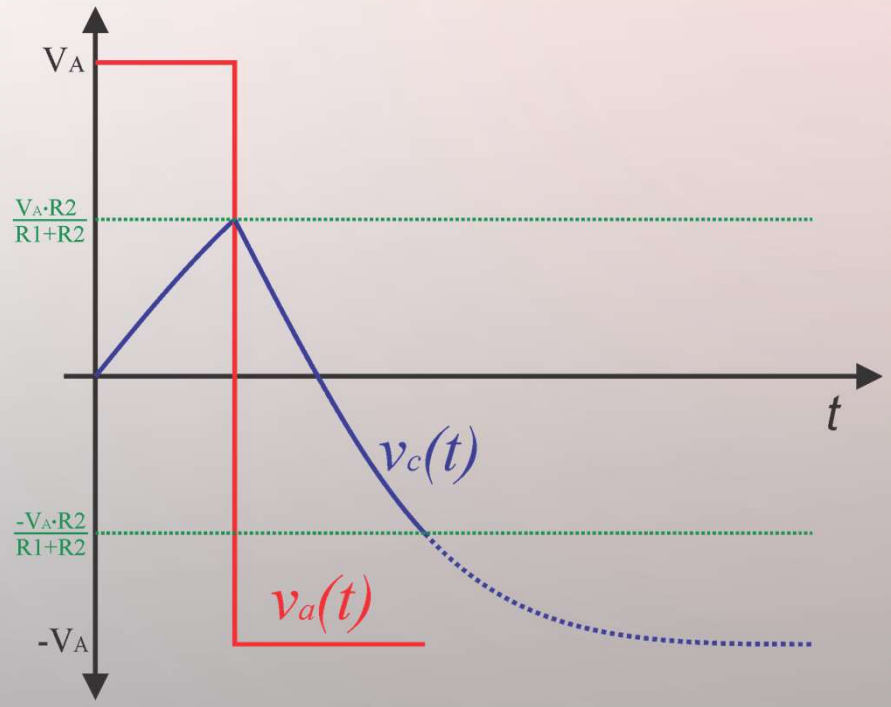




GENERADOR DE ONDA CUADRADA



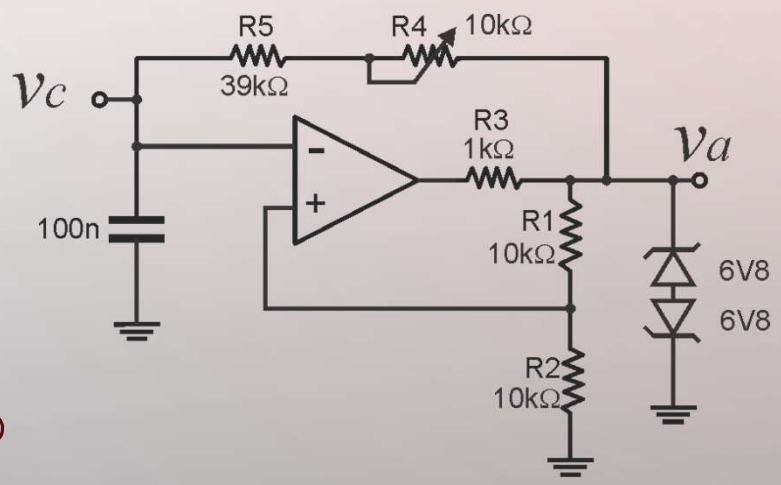
$$v_c(t) = (v_\infty) - (v_\infty - v_0) \cdot \mathrm{e}^{-\frac{t}{\tau}}$$

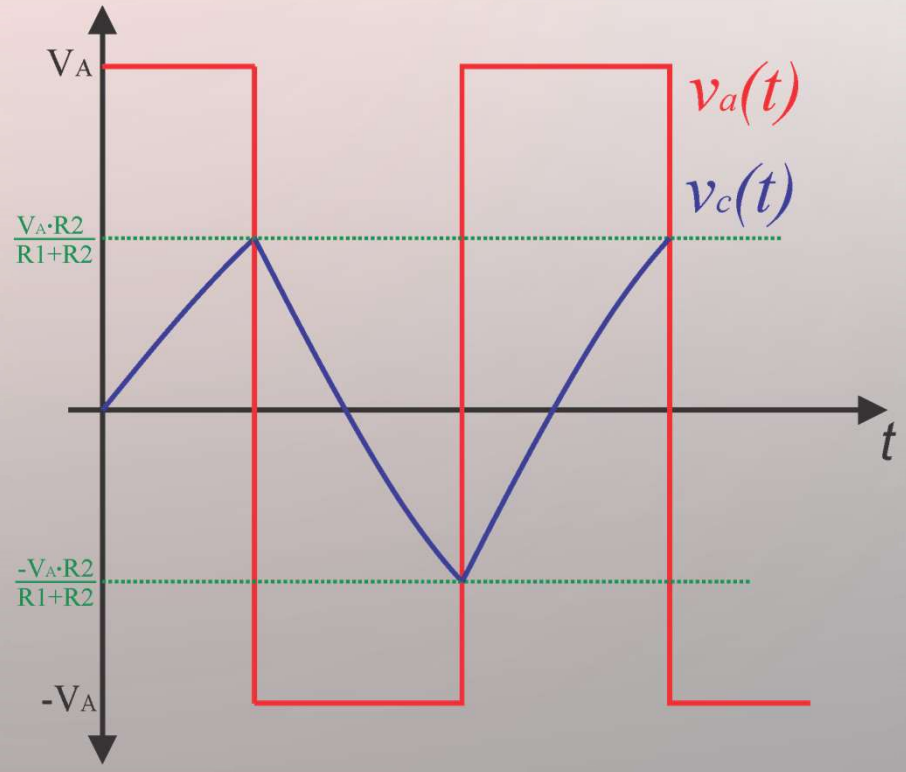
$$v_c(T/2) = -V_A - \left(-V_A - \frac{V_A}{2}\right) \cdot \mathrm{e}^{-\frac{T}{2\tau}} = -\frac{\mathrm{V_A}}{2}$$

$$T = -2\tau \cdot \ln\left(\frac{1}{3}\right) \quad \Rightarrow \quad \tau = 45{,}5\ ms$$

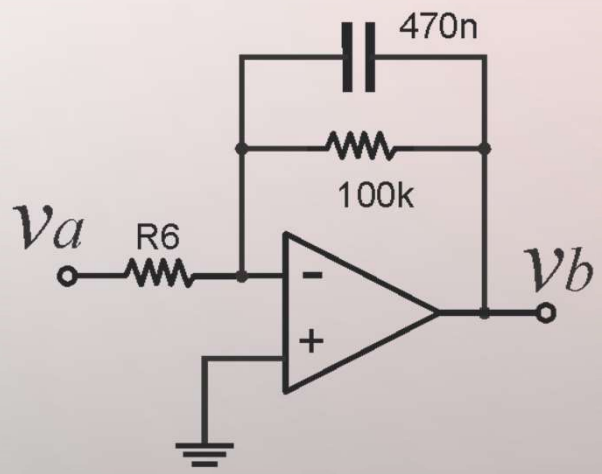
$$R4 + R5 = \frac{\tau}{100nF} = 45{,}5k\Omega$$

GENERADOR DE ONDA CUADRADA





INTEGRADOR



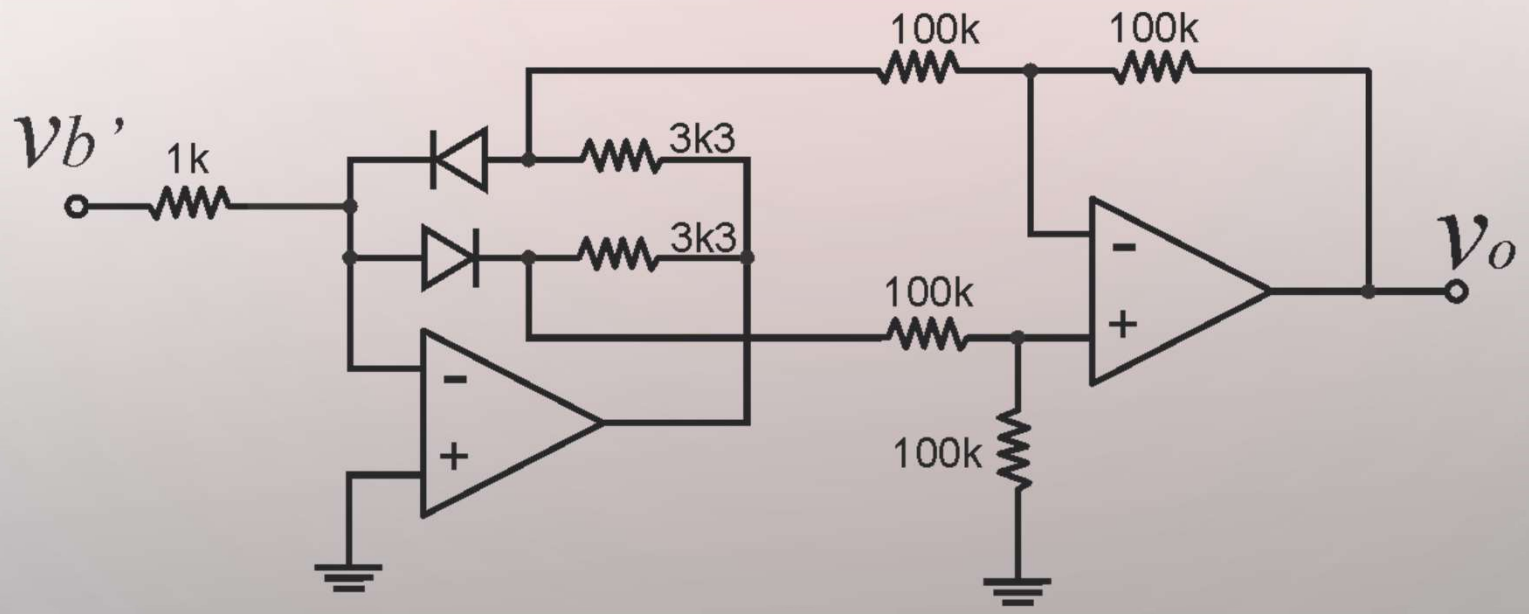
$$v_b(t) = v_b(t_0) + \int_{t_0}^{t} -\frac{v_a(t')}{R6} \cdot dt'$$

$$-2V = \frac{1}{470nF} \cdot \int_{0}^{\frac{T}{2}} -\frac{V_a}{R6} \cdot dt'$$

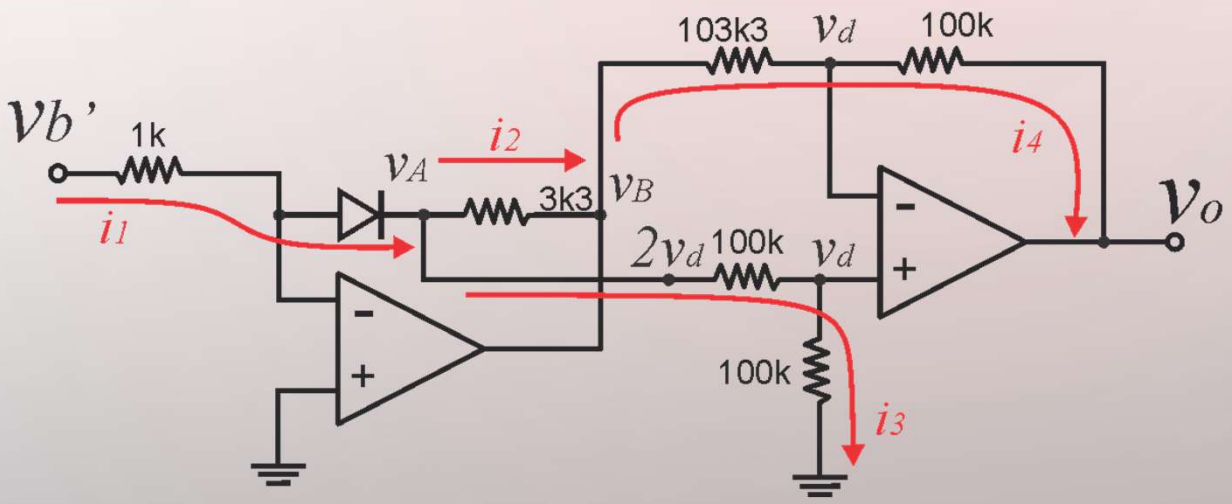
$$R6 = \frac{V_a \cdot T}{2 \cdot 2V \cdot 470nF} = 39{,}3k\Omega$$

Adoptamos: $R6 = 39k\Omega$

RECTIFICADOR



RECTIFICADOR



$v_b' > 0$

$i_1 = \frac{v_b'}{1k}$ $\quad i_3 = \frac{v_d}{100k}$ $\quad i_2 = i_1 - i_3$

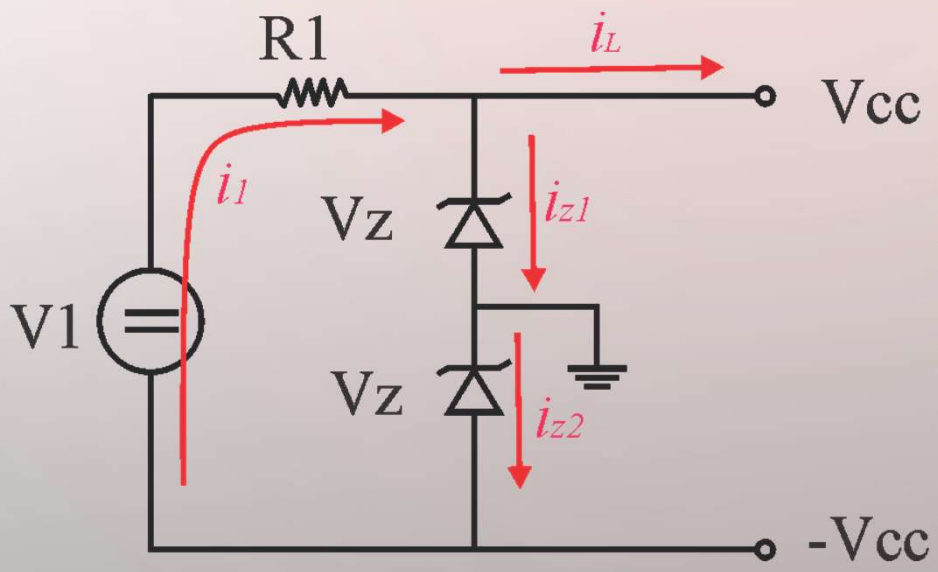
$v_A = 2v_D$ $\quad v_B = v_A - 3{,}3k \cdot i_2$

$i_4 = \frac{v_D - v_0}{100k}$ $\quad v_B - 103{,}3k \cdot i_4 = V_D$

$$v_o = v_b' \cdot \frac{100k}{103{,}3k} \cdot \frac{3{,}3k}{1k} = 3{,}2\ v_b'$$

Analogamente: $v_b' < 0 \rightarrow v_o = -3{,}2 \cdot v_b'$

FUENTE DOBLE CON ZENER



$$i_{Z_{MIN}} > 1mA$$

$$i_1 = i_L + i_Z \quad \rightarrow \quad i_1 = i_{Z_{MIN}} + i_{L_{MAX}}$$

$$i_1 = \frac{\mathrm{V1} - 2 \cdot \mathrm{Vcc}}{\mathrm{R1}}$$

$$\mathrm{R1} < \frac{\mathrm{V1} - 2 \cdot \mathrm{Vcc}}{i_{Z_{MIN}} + i_{L_{MAX}}}$$